

Conditional Probability

- The probability of an event B occurring when it is known that some event A has occurred is called the conditional probability and it is denoted by $P(B|A)$. This symbol is read as - the probability of B given A .

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0}$$

- In some cases, $P(B|A) = P(B)$, which means, the probability of B occurring has no dependency on being event A has occurred. When such a scenario is present, the events A & B are said to be independent. Two events A and B are independent iff;

$$\boxed{P(B|A) = P(B)} \quad \text{or}$$

$$\boxed{P(A|B) = P(A)}$$

here, we assume the existence of conditional probabilities. Otherwise, events A & B are dependent.

- By following the multiplicative rule or product rule, helps us to calculate the probability that two events will both occur. If in an experiment, the events A & B can both occur, then

$$\boxed{P(A \cap B) = P(A) P(B|A)}, \quad P(A) > 0$$

since, $(A \cap B)$ and $(B \cap A)$ are equivalent

$$P(A \cap B) = P(B \cap A) = P(B) P(A|B)$$

In other words, in this defⁿ it doesn't matter which is event A and which is event B .

- The two events A & B are independent, if.

$$\boxed{P(A \cap B) = P(A) P(B)}$$

Therefore, to obtain the probability that 2 independent events will both occur, we simply find the product of their individual probabilities.

- The multiplicative rule can be extended to more than 2-event situations.

If in an experiment, events $A_1, \dots, A_k$ can occur then,

$$\boxed{P(A_1 \cap \dots \cap A_k) = P(A_1) P(A_2|A_1) \dots P(A_k|A_1 \cap \dots \cap A_{k-1})}$$

- If these events are independent, then

$$P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1) P(A_2) \dots P(A_k)$$

- A collection of events $A = \{A_1, \dots, A_n\}$ are mutually independent if for any subset of A , $A_{i_1}, \dots, A_{i_k}$, $k \leq n$, we have

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k})$$

$$P(A \cap B) = P(A)P(B) \quad \text{if } A \text{ and } B \text{ are independent}$$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

$$P(A \cap B) = P(A)P(B)$$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$

if A and B are independent, then $P(A \cap B) = P(A)P(B)$